

WEEK 2: Product and Category Management System

-S.KAVIYA (E25AI119)

Scenario:

The Art Work Shop wants to maintain information about products and their categories. Each product belongs to a specific category, and the system should manage product details such as name, price, and stock availability.

1. Product and Category Tables:

Category table:

```
SQL> CREATE TABLE Category(Category_ID VARCHAR2(7),Category_Name VARCHAR2(20),Description VARCHAR(50));  
Table created.
```

Product table:

```
SQL> CREATE TABLE Product(Product_ID VARCHAR2(7),Product_Name VARCHAR2(20),Category_ID VARCHAR2(7),Price NUMBER(9,2),Stock_Quantity Number);  
Table created.
```

Product and Category tables are created

2. Define Relationships and Constraints

Relationships:

One Category can have many Products.

The relationship between Category and Product is One to Many (1 → *).

Primary Key:

In Category Table, Category_ID uniquely identifies each category.

In Product Table, Product_ID uniquely identifies each product.

Foreign Key:

Category_ID acts as Foreign Key in the Product Table.

Other Constraints:

NOT NULL: Product name and category name cannot be empty.

Example: The Product_Name and Category_Name fields should not be left empty because they are required to identify the records.

CHECK: Product price must always be a positive number.

Example: The product price must be greater than zero, so an invalid negative price cannot be entered.

UNIQUE: No two categories can have the same category name.

Example: Every category should have a different Category_Name. Two categories cannot be created with the same name.

Defining relationship and create constraints for product and category:

Primary key:

```
SQL> Alter Table Category Add Constraint PK_Category PRIMARY KEY (Category_ID);
Table altered.

SQL> Alter Table Product Add Constraint PK_Product PRIMARY KEY (Product_ID);
Table altered.
```

Creating relationship using Foreign key:

```
SQL> Alter Table Product
  2 Add Constraint FK_Product_Category
  3 FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID);
Table altered.
```

Adding constraints

not null:

```
SQL> Alter Table Category
  2 MODIFY Category_Name VARCHAR(100) NOT NULL;
Table altered.
```

```
SQL> Alter Table Product
  2 MODIFY Product_Name VARCHAR(150) NOT NULL;
Table altered.
```

Check:

```
SQL> Alter Table Product Add Constraint CHK_Price CHECK (Price > 0);
Table altered.
```

```
SQL> Alter Table product Add Constraint CHK_Stock CHECK (Stock_Quantity >=0);  
Table altered.
```

Unique:

```
SQL> Alter Table Category Add Constraint UQ_Category_Name UNIQUE (Category_Name);  
Table altered.
```

3.Inserting values :

Category Table:

```
SQL> INSERT INTO Category  
2 VALUES(  
3 '&Category_ID',  
4 '&Category_Name',  
5 '&Description');  
Enter value for category_id: SEM0001  
old 3: '&Category_ID',  
new 3: 'SEM0001',  
Enter value for category_name: Electronics  
old 4: '&Category_Name',  
new 4: 'Electronics',  
Enter value for description: Electronics Gadgets and Devices  
old 5: '&Description')  
new 5: 'Electronics Gadgets and Devices')  
  
1 row created.  
  
SQL> /  
Enter value for category_id: SEM0002  
old 3: '&Category_ID',  
new 3: 'SEM0002',  
Enter value for category_name: Clothing  
old 4: '&Category_Name',  
new 4: 'Clothing',  
Enter value for description: Men , Women and Kids Apparels  
old 5: '&Description')  
new 5: 'Men , Women and Kids Apparels')  
  
1 row created.
```

```
SQL> /
Enter value for category_id: SEM0003
old   3: '&Category_ID',
new   3: 'SEM0003',
Enter value for category_name: Stationary
old   4: '&Category_Name',
new   4: 'Stationary',
Enter value for description: Notebooks,Pens and Papers
old   5: '&Description')
new   5: 'Notebooks,Pens and Papers')

1 row created.

SQL> /
Enter value for category_id: SEM0004
old   3: '&Category_ID',
new   3: 'SEM0004',
Enter value for category_name: Home Appliances
old   4: '&Category_Name',
new   4: 'Home Appliances',
Enter value for description: Home essentials and needs
old   5: '&Description')
new   5: 'Home essentials and needs')

1 row created.
```

```
SQL> /
Enter value for category_id: SEM0005
old   3: '&Category_ID',
new   3: 'SEM0005',
Enter value for category_name: Mobile Parts
old   4: '&Category_Name',
new   4: 'Mobile Parts',
Enter value for description: New Spares for Mobile
old   5: '&Description')
new   5: 'New Spares for Mobile')

1 row created.

SQL> /
Enter value for category_id: SEM0006
old   3: '&Category_ID',
new   3: 'SEM0006',
Enter value for category_name: Provisions
old   4: '&Category_Name',
new   4: 'Provisions',
Enter value for description: Kitchen Needs
old   5: '&Description')
new   5: 'Kitchen Needs')

1 row created.
```

Product table:

```
SQL> INSERT INTO Product
  2 VALUES(
  3 '&Product_ID',
  4 '&Product_Name',
  5 '&Category_ID',
  6 &Price,
  7 &Stock_Quantity);
Enter value for product_id: P1001
old 3: '&Product_ID',
new 3: 'P1001',
Enter value for product_name: Laptops
old 4: '&Product_Name',
new 4: 'Laptops',
Enter value for category_id: SEM0001
old 5: '&Category_ID',
new 5: 'SEM0001',
Enter value for price: 67500
old 6: &Price,
new 6: 67500,
Enter value for stock_quantity: 2
old 7: &Stock_Quantity)
new 7: 2)

1 row created.
```

```
SQL> /
Enter value for product_id: P1002
old 3: '&Product_ID',
new 3: 'P1002',
Enter value for product_name: Croptops and shirts
old 4: '&Product_Name',
new 4: 'Croptops and shirts',
Enter value for category_id: SEM0002
old 5: '&Category_ID',
new 5: 'SEM0002',
Enter value for price: 45000
old 6: &Price,
new 6: 45000,
Enter value for stock_quantity: 456
old 7: &Stock_Quantity)
new 7: 456)

1 row created.
```

```
SQL> /
Enter value for product_id: P1003
old 3: '&Product_ID',
new 3: 'P1003',
Enter value for product_name: Fish tank
old 4: '&Product_Name',
new 4: 'Fish tank',
Enter value for category_id: SEM0003
old 5: '&Category_ID',
new 5: 'SEM0003',
Enter value for price: 3400
old 6: &Price,
new 6: 3400,
Enter value for stock_quantity: 4
old 7: &Stock_Quantity)
new 7: 4)

1 row created.
```

```

SQL> /
Enter value for product_id: P1004
old 3: '&Product_ID',
new 3: 'P1004',
Enter value for product_name: Television
old 4: '&Product_Name',
new 4: 'Television',
Enter value for category_id: SEM0004
old 5: '&Category_ID',
new 5: 'SEM0004',
Enter value for price: 75000
old 6: &Price,
new 6: 75000,
Enter value for stock_quantity: 5
old 7: &Stock_Quantity)
new 7: 5)

1 row created.

```

```

SQL> /
Enter value for product_id: P1005
old 3: '&Product_ID',
new 3: 'P1005',
Enter value for product_name: Unruled notes
old 4: '&Product_Name',
new 4: 'Unruled notes',
Enter value for category_id: SEM0005
old 5: '&Category_ID',
new 5: 'SEM0005',
Enter value for price: 7500
old 6: &Price,
new 6: 7500,
Enter value for stock_quantity: 765
old 7: &Stock_Quantity)
new 7: 765)

1 row created.

```

```

SQL> /
Enter value for product_id: P1006
old 3: '&Product_ID',
new 3: 'P1006',
Enter value for product_name: Tiles
old 4: '&Product_Name',
new 4: 'Tiles',
Enter value for category_id: SEM0006
old 5: '&Category_ID',
new 5: 'SEM0006',
Enter value for price: 55000
old 6: &Price,
new 6: 55000,
Enter value for stock_quantity: 670
old 7: &Stock_Quantity)
new 7: 670)

1 row created.

```

After insertion category table:

```

1 row created.

SQL> Select * from Category;

CATEGOR
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
SEM0001
Electronics
Electronics Gadgets and Devices

SEM0002
Clothing
Men , Women and Kids Apparels

CATEGOR
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
SEM0003
Stationary
Notebooks,Pens and Papers

SEM0004
Home Appliances

CATEGOR
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
Home essentials and needs

```

```

SEM0005
Mobile Parts
New Spares for Mobile

SEM0006

CATEGOR
-----
CATEGORY_NAME
-----
DESCRIPTION
-----
Provisions
Kitchen Needs

6 rows selected.

```

After insertion product table:

```
SQL> Select * from Product;
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
P1001
```

```
Laptops
```

```
SEM0001      67500          2
```

```
P1002
```

```
Croptops and shirts
```

```
SEM0002      45000         456
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
P1003
```

```
Fish tank
```

```
SEM0003      3400          4
```

```
P1004
```

```
Television
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
SEM0004      75000          5
```

```
P1005
```

```
Unruled notes
```

```
SEM0005      7500         765
```

```
P1006
```

```
PRODUCT
```

```
-----  
PRODUCT_NAME
```

```
-----  
CATEGOR      PRICE STOCK_QUANTITY
```

```
-----  
Tiles
```

```
SEM0006      55000         670
```

```
6 rows selected.
```

CRUD OPERATIONS:

Insertion record:

```
SQL> INSERT INTO Product (Product_ID,Product_Name,Category_ID,Price,Stock_Quantity)
  2  VALUES ('P1007','Tablet','SEM0002',10000,10);

1 row created.
```

Update price:

```
SQL> UPDATE Product SET Price = 76000 WHERE Product_Name = 'Tiles';

1 row updated.
```

Delete a product:

```
SQL> DELETE FROM Product WHERE Product_Name = 'Tiles';

1 row deleted.
```

4. Category-Wise Product Reports:

SQL query:

```
SELECT c.CATEGORY_ID,
       c.CATEGORY_NAME,
       p.PRODUCT_ID,
       p.PRODUCT_NAME,
       p.PRICE,
       p.STOCK_QUANTITY
FROM product p
JOIN Category c
ON  p.CATEGORY_ID = c.CATEGORY_ID;
```

```
SQL> SELECT c.Category_Name,p.Product_Name,p.Price,p.Stock_Quantity
2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID;
```

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Electronics

Laptops

67500 2

Clothing

Croptops and shirts

45000 456

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Stationary

Fish tank

3400 4

Home Appliances

Television

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

75000 5

Mobile Parts

Unruled notes

7500 765

Clothing

CATEGORY_NAME

PRODUCT_NAME

PRICE STOCK_QUANTITY

Tablet

10000 10

6 rows selected.

Counting Number of products in each Category:

```
SQL> SELECT c.Category_Name, COUNT(p.Product_ID) AS Product_Count
  2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
  3 GROUP BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Clothing 2

Stationary 1

Home Appliances 1

CATEGORY_NAME

PRODUCT_COUNT

Mobile Parts 1

Electronics 1

Finding highest priced product in each category:

```
SQL> SELECT c.Category_Name, p.Product_Name, p.Price AS Highest_Price FROM Product p
  2 JOIN Category c ON p.Category_ID = c.Category_ID
  3 WHERE p.Price = (
  4     SELECT MAX(p2.Price)
  5     FROM Product p2
  6     WHERE p2.Category_ID = p.Category_ID
  7 );
```

CATEGORY_NAME

PRODUCT_NAME

HIGHEST_PRICE

Electronics
Laptops 67500

Clothing
Croptops and shirts 45000

CATEGORY_NAME

PRODUCT_NAME

HIGHEST_PRICE

Stationary
Fish tank 3400

Home Appliances
Television

CATEGORY_NAME

```

CATEGORY_NAME
-----
PRODUCT_NAME
-----
HIGHEST_PRICE
-----
      75000
Mobile Parts
Unruled notes
      7500

```

Display categories having more than 5 products

```

SQL> SELECT c.Category_Name
      2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
      3 GROUP BY c.Category_Name HAVING COUNT(p.Product_ID) > 5;

no rows selected

```

No rows selected because all categories having less than 5 products.

Find average product price category-wise

```

SQL> SELECT c.Category_Name, AVG(p.Price) AS Avg_Price
      2 FROM Product p JOIN Category c ON p.Category_ID = c.Category_ID
      3 GROUP BY c.Category_Name;

CATEGORY_NAME
-----
AVG_PRICE
-----
Clothing
      27500

Stationary
      3400

Home Appliances
      75000

CATEGORY_NAME
-----
AVG_PRICE
-----
Mobile Parts
      7500

Electronics
      67500

```